

Emerging Technologies: The Internet of Things

DICT422

Practical 1: Applying Five-Layer Architecture to a Real-World Solution

Due Date: Monday, 17th August

Presentations: Tuesday, 18th August

Instructions:

Work in groups of 5.

Required:

Identify a real-world challenge in your community or any surrounding environment, and propose an IoT-based solution applying the five-layer IoT architecture.

The five-layer IoT architecture must show a clear flow of information.

Guide your solutions based on the tasks in the table below.

Tasks	Marks
1. Introduction	[10]
1.1 Introduce the problem	2
1.2 Describe the problem	8
2. Proposed solution	[20]
2.1 Explain why IoT is relevant to provide a solution	5
2.2 Explain how IoT will solve or improve the challenge	5
2.3 Identify and discuss five IoT challenges related to the	10

proposed solution	
3. IoT Architectural Design	[40]
3.1 Identify and describe the Five Layer IoT Architecture	15
3.2 Provide an illustrative five-layer architectural diagram aligned to your identified problem	25
4. Participation of each member	[5]
5. Presentation	[25]
5.1 Illustrate understanding of the topic	5
5.2 Presenters' preparedness (Confidence, audible, fewer ummms, uhhhs)	5
5.3 Slides readability (layout, font types and sizes used are consistent and professional)	5
5.4 Creativity: Well-designed slides with animation, videos, and transitions	5
5.5 Time management (10 minutes)	2
5.6 Presenters are able to answer questions	3
Total	100

Report: Submit in Word format.

Must be justified, 1.5 line spacing, Times New Roman font style, page numbered, in-text referencing, and a list of references

Bonus marks: 5 marks, if all report requirements are addressed.

Prepared by: Dr M Chere